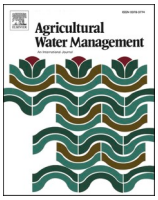
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Review



A global meta-analysis of yield and water use efficiency of crops, vegetables and fruits under full, deficit and alternate partial root-zone irrigation

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ABSTRACT

Deficit irrigation (DI) and alternate partial root-zone irrigation (APRI) are widely recognized water-saving irrigation methods. This meta-analysis provided a systematic and quantitative result of the relative changes in yield and water use efficiency (WUE) of seven agricultural products (maize, wheat, cotton, potato, tomato, apple and grape) under conventional full irrigation (FULL), APRI and DI globally. Three comparisons were performed: (1) APRI by FULL, (2) APRI by DI and (3) DI by FULL. The results showed that the relative changes of yield and WUE differed among the three comparisons. Higher yield and WUE were achieved in fruits (apple and grape) under APRI compared with FULL or DI. However, the percentage changes in yield and WUE under APRI were not significantly different in vegetables (potato and tomato) and crops (maize, wheat and cotton). Compared with FULL, APRI performed better for acidic soils, clay loam or sandy loam, high bulk density ($\geq$ 1.35 g cm^{-3}), irrigation amount of 80–100% FULL, and was more appropriate for furrow irrigation in drier and colder environments. Compared with FULL, DI performed better for clay loam, low bulk density ($<$ 1.35 g cm^{-3}) and precipitation $>$ 200 mm. Yields under APRI and DI were remarkably lower than those under FULL, while the corresponding WUE were significantly higher. Compared with DI, fruits with furrow irrigation and in soils with clay and sandy loam were better under APRI. There was no significant difference in yield between APRI and DI, but WUE under APRI was higher. APRI increased both yield and WUE in regions with low average annual temperature ($<$ 9 ℃). It is critical to weigh the cost of yield penalty, the economic benefit of saving water and the cost of implementing irrigation method before choosing an effective water-saving irrigation method.

1. Introduction

The global climate change, water resources shortage and ever-growing population have brought great challenges to the sustainable development of agriculture around the world (Hanjra and Qureshi, 2010; Wheeler and von Braun, 2013). By 2050, the global population will reach 9 billion, and the demand for foods will double (Bagatur, 2014). Water is a major factor influencing agricultural production, especially in arid and semi-arid regions (Solomon and Cramer, 1993; Zheng et al., 2018, 2021; FAO, 2012). Sufficient availability of water in the root zone is crucial throughout the growing season of crops (Xu et al., 2020; Yang et al., 2006; Yawson et al., 2014). Therefore, it is of significance to find ways to reduce agricultural water use while maintaining or even increasing crop yields (Rosegrant et al., 2009; Yan et al., 2019).

Deficit irrigation (DI) and alternate partial root-zone irrigation (APRI) are two widely recognized effective water-saving irrigation methods. DI involves the application of water in a strictly controlled manner that is lower than the total water demand during the specified part or the seasonal cycle of crop development (Liu et al., 2020; Kriedemann and Goodwon, 2009). By contrast, APRI alternates irrigation in space or time to create a dry and wet cycle in different parts of the root system to maintain plant water conditions under maximum water potential, and to control the vegetative growth (Kriedemann and Goodwon, 2009; Sadras, 2009). Although there are many differences between the two irrigation methods, their purpose is to directly reduce the input of irrigation water in time or space, so as to improve WUE of crops without significantly reducing crop yields compared to the conventional full irrigation (Kirda et al., 2007; Kriedemann and Goodwin, 2003; Kriedemann and Goodwon, 2009; Morison et al., 2008; Sadras, 2009).

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Since APRI was proposed, it has received attention worldwide (Pérez-Pérez et al., 2020) and a large number of related experiments have been conducted. The results showed that APRI can save 30–50% of irrigation water and increase WUE, while reducing crop yields slightly in many crops, such as oat (Lin et al., 2012), grape (Intrigliolo and Castel, 2009; Kusakabe et al., 2016), cotton (Tang et al., 2005, 2010), papaya (Santos et al., 2021), maize (Fu et al., 2017) and tomato (Sarker et al., 2016). However, the results of some studies were inconsistent and concluded that APRI can increase the crop yields. For example, Du et al. (2017) reported that APRI with eight irrigation times significantly increased the yield and WUE of apple in arid northwest China. Du et al. (2008a) found that the seed cotton yield under alternate drip irrigation (ADI) was 5.61% higher than that under conventional drip irrigation (CDI). Sadras (2009) concluded that the yield produced by APRI and conventional irrigation was within a range of ± 20% in 80% of cases, and APRI outperformed deficit irrigation by 20% or more in other cases. Adu et al. (2018) found the yields of different crops produced by deficit irrigation and APRI were similar. Obviously, the conclusions on the crop yield and WUE among these three irrigation methods (DI, APRI and FULL) are still inconsistent.

In view of this situation, meta-analysis can provide a solution. Meta-analysis is a statistical model used to quantitatively analyze and summarize the results from multiple independent studies with the same theme, so that meta-analysis researchers can draw conclusions from the concatenated dataset (Stanley and Jarrell, 1989). In present, meta-analysis has been widely used in agricultural studies. Some meta-analyses have also been done on the impact of different irrigation methods on crop yield and WUE. For example, Yu et al. (2020) applied a meta-analysis to quantify the effects of deficit irrigation on water use efficiency and yield of wheat relative to full irrigation. Adu et al. (2018) employed a meta-analysis to study the influence of full irrigation, deficit irrigation and partial root-zone irrigation on crop yields. Carrijo et al. (2017) conducted a meta-analysis to quantify the effect of alternate wetting and drying on rice yield and water use. However, the previous meta-analyses on various irrigation methods only focused on the impact of different factors on various indicators of a single crop or on a single indicator of various crops. There is a lack of a systematic and quantitative analysis of the yield and WUE of various crops under DI, APRI and FULL on a global scale.

Therefore, a global meta-analysis was distinctly conducted to examine the relative changes of crop yield and WUE under APRI by FULL, APRI by DI and DI by FULL. In order to enable comparisons under similar conditions as much as possible, some of the related crops, soil and irrigation management practices contributing to the differences in crop yield and WUE were considered.

2. Materials and methods

2.1. Data collection

Relevant peer-reviewed studies from 1997 to 2020 were collected. The search engines used in this article included Google Scholar (https://xueshu.soogle.top/), Elsevier (Science Direct) (https://www.sciencedirect.com/search) and China National Knowledge Infrastructure (https://www.cnki.net/) using the key words: “partial root drying” OR “deficit irrigation” OR “partial root-zone” OR “irrigation strategies/regimes” OR “water-saving irrigation” AND “water use efficiency” AND “yield”. In total, there were 90 publications, including 305 pairs of observations of yield and WUE that satisfied our criteria for the meta-analysis. The list of the 90 publications from which the data were extracted are listed in Appendix A.

These publications were screened using the following criteria. (i) Only publications describing experiments conducted in the field with side-by-side comparisons of FULL or DI and APRI were selected and pot or rain shelter were excluded. (ii) Yield or WUE data in APRI and FULL or DI treatments were reported. It should be noted that, in this meta-analysis, we focused on WUE at field level (the ratio of yield divided by water consumption (ET)) (Viets, 1962). WUE at leaf or biomass level were excluded. (iii) The primary data of yield or WUE under APRI and FULL or DI must be comparable. (iv) The mean, sample size and a measure of dispersion (SE or SD; not necessarily mandatory) as numerical or graphical data were reported, or SD of yield or WUE could be calculated from the reported data for the three irrigation methods. When data were presented graphically, the figures were digitized to extract the numerical values using the Get-Data Graph Digitizer (ver. 2.22, Russian Federation). (v) More than two years of data were reported.

After that, we classified and stored these data (as described in Appendix A) by these criteria: (i) agricultural species (maize, wheat, cotton, potato, tomato, apple and grape); (ii) soil texture (clay, sandy clay, silty clay loam, clay loam, sandy clay loam, silt loam, loam, sandy loam and sand according to the USDA soil classification) (USDA, 1993); (iii) soil bulk density (< 1.35 g cm^{-3} and ≥ 1.35 g cm^{-3}) using by USDA soil texture classes (USDA, 1993); (iv) soil pH (pH < 7 and ≥ 7); (v) irrigation method (sprinkler, drip and furrow irrigation); (vi) the period when APRI or DI was undertaken (reproductive stage and throughout the season); (vii) the amount of water, as a proportion of the amount of FULL or DI, (80–100%, 60–80%, 40–60% and $< 40\%$).

2.2. Statistical analysis

We extracted mean and standard error (SE) of each treatment, from which the standard deviation (SD) is calculated as (Hedges et al., 1999; Luo et al., 2006):

$$SD = SE\sqrt{n} \tag{1}$$

where n is the sample size.

The means in the treatment (Xe) and control group (Xc) are used to compute a response ratio:

$$\ln RR = \ln\left(\frac{Xe}{Xc}\right) \tag{2}$$

In every study, Xe and Xc are the mean values of either WUE or yield for the experiment treatment and control treatment, respectively; lnRR is a unit-free index.

In addition, the variance of lnRR (v) is calculated as (Luo et al., 2006):

$$v = \frac{sd_1^2}{n_1 \times Xe^2} + \frac{sd_0^2}{n_0 \times Xc^2} \tag{3}$$

where sd_1 and sd_0 are the standard deviations for the experiment treatment and control treatment, respectively; n_1 and n_0 are the sample sizes of the experiment treatment and control treatment, respectively.

Usually, the weighted mean is used to produce the greatest precision since statistical precision of each experiment is different (Curtis and Wang, 1998). The weighted mean response ratios ($lnRR_+$) and weight are calculated as follows (Luo et al., 2006):

$$\ln RR_+ = \frac{\sum_{i=1}^{k} w_i lnRR}{\sum_{i=1}^{k} w_i} \tag{4}$$

$$w_i = \frac{1}{v} \tag{5}$$

where k is the study number, w_i is the weight of the effect size of the ith experiment group. The variance of $lnRR_+$ (Var) is calculated as follows:

$$Var = \frac{1}{\sum_{i=1}^{k} w_i} \tag{6}$$

The 95% confidence intervals of $\ln RR_+$ (95%CI) are calculated as followed:

$$95\%CI = \ln RR_+ \pm 1.96 \times \sqrt{Var} \quad (7)$$

A random-effect model was adopted in this meta-analysis to examine the performance of yield and WUE under APRI (compared with FULL or DI) or DI (compared with FULL). For better understanding, the results of this meta-analysis were all reported as the percentage change for the experimental treatment relative to the control treatment ((RR_+-1) × 100%).

2.3. Publication bias

The funnel plot can be used to intuitively assess whether there is a publication bias. If there is no publication bias, the funnel plot should be similar to an inverted symmetric funnel; however, if there is publication bias, the plot will be an asymmetric funnel (Adu et al., 2018; Duval and Tweedie, 2000; Ferreira et al., 2016; Higgins and Green, 2011). Hereafter, we need to further evaluate its potential impact on the overall effect size through the "trim and fill" method (Duval and Tweedie, 2000; Ferreira et al., 2016; Jennions et al., 2013). According to the results of funnel plot, the yield of APRI by FULL was near symmetrical, and the trim and fill analysis indicated that there was no missing study. However, for the funnel plots of yield of APRI by DI, DI by FULL and WUE of APRI by FULL, APRI by DI, DI by FULL, there were five, eleven, three, two and two estimated missing studies, respectively. Although it changed the magnitude of the effect size, it did not change the significance. Meanwhile, the Rosenthal's fail-safe number (N*fs*) (Orwin, 1983) for estimating the number of additional studies needs to invalidate the statistically significant results: $Nfs > 5 \times k + 10$, where k is the number of studies. The *Nfs* values of yield and WUE calculated using Stata were 57,361 and 1768, respectively. The two *Nfs* were significantly higher than $1175(5 \times 231 + 10)$ and $375(5 \times 73 + 10)$, so the publication bias was not a big problem for this meta-analysis.

2.4. Data analyses

In this meta-analysis, all data analyses were conducted by the Stata/SE 15.1 software (Stata, College Station, TX, USA). The mapping software used in this meta-analysis included Origin v9.0 and GraphPad Prism8.

3. Results

3.1. Overview of the dataset

The database was selected from 90 articles reporting the studies of seven agricultural species from 20 countries around the world. The studied species included crops (maize, wheat and cotton), vegetables (potato and tomato) and fruits (apple and grape). The relative frequency distribution of effect sizes on yield and WUE followed Gaussian normal

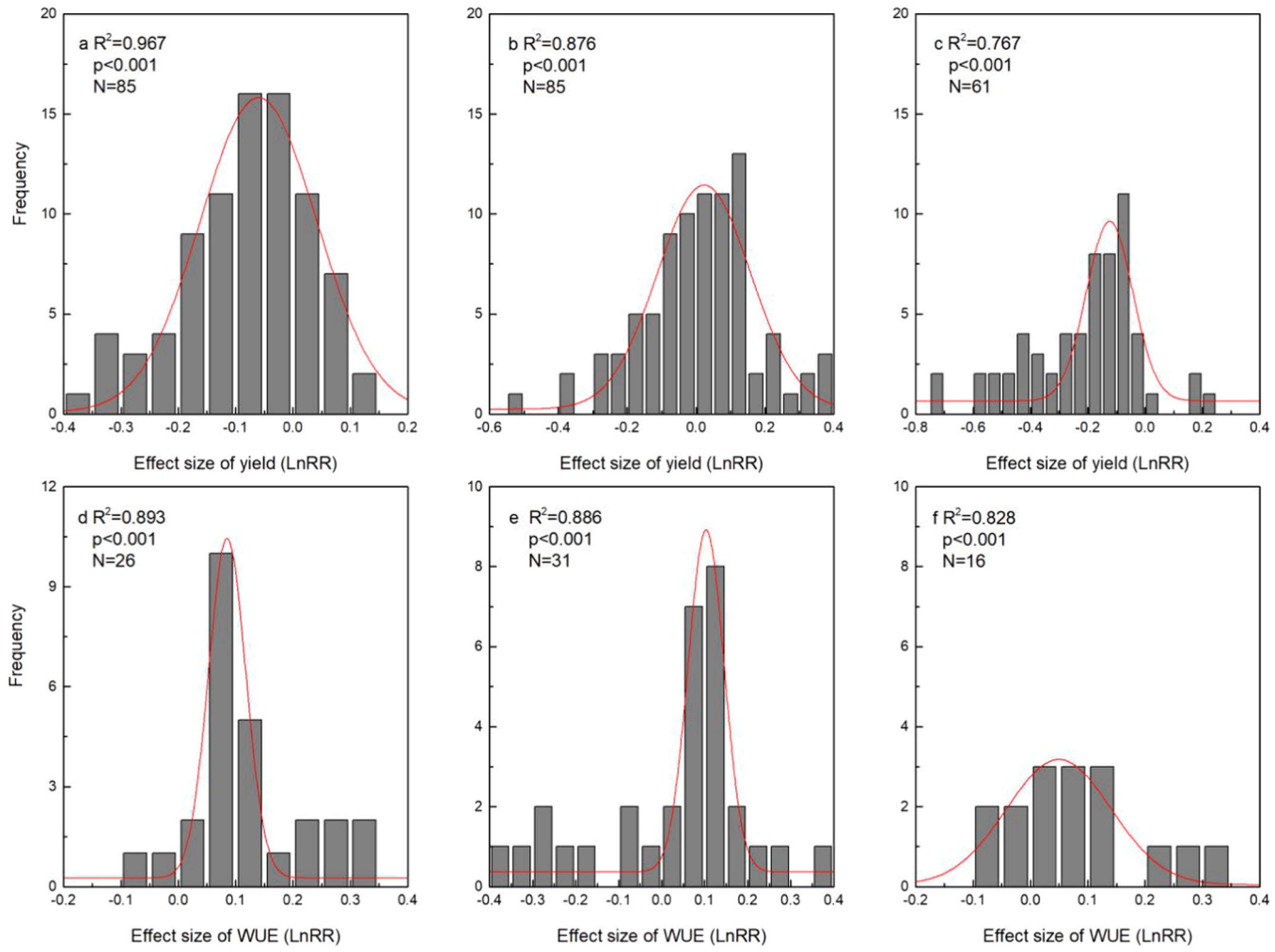


Fig. 1. Frequency distributions of effect sizes for yield and WUE responding to different irrigation methods (APRI-to-FULL (a, d), APRI-to-DI (b, e), DI-to-FULL (c, f)). The lines are fitted normal (GaussAmp) distribution to frequency data sets and N is sample size. The p indicates the significant level.

distributions (Fig. 1), indicating that the dataset were homogeneous (Shan and Yan, 2013).

Yields were significantly reduced under APRI and DI compared with FULL, and the overall effect sizes of yield for APRI by FULL and DI by FULL were −7.09% (95%CI: −7.96% to −6.21%) and −17.44% (95%CI: −18.8% to −16%). There was no significant difference between APRI and DI, and the overall effect size of yield was −1.54% (95%CI: −2.79–0.28%) (Fig. 2). As for WUE, the overall effect size of these three comparisons were all significantly increased, and WUE for APRI by FULL, APRI by DI and DI by FULL were 8.46% (95%CI: 6.81–10.12%), 4.77% (95%CI: 3.03–6.55%) and 12.78% (95%CI: 8.47–17.26%) (Fig. 2).

3.2. Yield and WUE response of agricultural species groups to different irrigation methods

3.2.1. APRI by FULL

Compared with FULL, APRI significantly reduced yield and increased WUE for all agricultural species groups (Fig. 3a, d). APRI had the smallest negative effect on yield of fruits (−6.72%), while crops had the largest negative effect on yield (−7.18%), but there were no significant difference among the agricultural species groups (Fig. 3a). However, the WUE of fruits was increased most by 22.33% (95%CI: 14.02–31.25%), which was significantly higher than that of crops (7%) (Fig. 3d). Obviously, the effect of APRI was best for fruits and worst for crops.

3.2.2. APRI by DI

Compared with DI, yield and WUE of APRI showed significantly different trends among the agricultural species groups. The yield and WUE of fruits both increased, while vegetables showed opposite trends, but the WUE of them were not significantly different (Fig. 3b, e). Among them, the yield of fruits showed an increased trend by 4.46% (95%CI: 0.89–8.16%), while both vegetables and fruits showed decreased trend by −8.64 (95%CI: −10.24% to −7.00%) and −3.97% (95%CI: −5.53% to −2.38%), respectively (Fig. 3b). Apart from crops (5.77%), the changes in WUE were not significant (1.05% and −0.76%, respectively) for fruits and vegetables (Fig. 3e).

3.2.3. DI and FULL

DI decreased yield significantly but increased WUE relative to FULL among different agricultural species groups, with varying degrees of success (Fig. 3c, f). Crops reduced yield least by −10.03% (95%CI: −12.62% to −7.37%) while its WUE increased least (6.96%). As for fruits and vegetables, there was no significant difference between the reduction in yield and the increase in WUE.

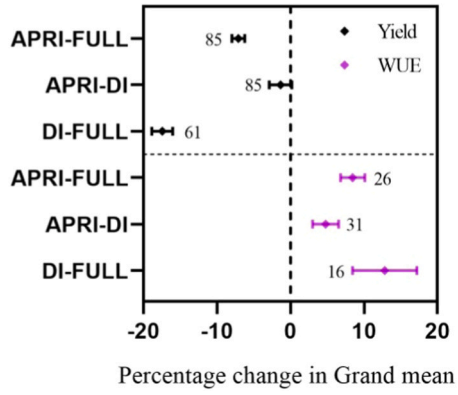


Fig. 2. The effects of different irrigation methods on yield and WUE. Mean effect and 95% confidence intervals are shown. Mean effect size = 0 (vertical dotted line) indicated no effect, mean effect > 0 indicates yield or WUE advantage of APRI over FULL, APRI over DI, DI over FULL and mean effect size < 0 indicates yield or WUE advantage of FULL over APRI, DI over APRI, FULL over DI. Number beside the every bar represents the sample size.

3.3. Yield and WUE response of irrigation management to different irrigation methods

3.3.1. APRI by FULL

The irrigation methods decreased yield and increased WUE under APRI, with varying degrees of success (except sprinkler irrigation) (Fig. 4a, b). The effect size for yield in response to sprinkler irrigation was not significant, which may be due to the small sample size (n = 4) (Fig. 4a). From all observations, APRI decreased yield by 1.11%, 8.11% and 4.78% under sprinkler, drip and furrow irrigation, but it significantly increased WUE by 8.36% and 8.89% under drip and furrow irrigation, respectively. Comprehensively considering yield and WUE, furrow irrigation performed best (Fig. 4a, b).

In term of the deficit stage, APRI (compared to FULL) significantly decreased yield by 8.14% (reproductive stage) and 6.87% (throughout the season), but it significantly increased WUE by 11.66% and 8.17%, respectively. However, the differences of yield and WUE between these two stages were not significant (Fig. 4c, d).

The effects of APRI on yield and WUE also varied with deficit proportions, expressed as a fraction of FULL (Fig. 4e, f), and the degree of APRI on yield was positively correlated with deficit proportions. When the deficit proportion ranged 80–100%, the reduction of yield was minimal (−3.16%, 95%CI: −4.82% to −1.46%). Moreover, APRI decreased yield by 8.09%, 2.90% and 25.37% when the deficit proportions were within the range of 60–80%, 40–60% and < 40%, respectively (Fig. 4e). The percentage change in WUE showed a decreasing and then increasing trend as the deficit proportion decreased, with the smallest WUE of 4.56% (95%CI: 1.38–7.84%) (irrigation proportion was 60–80% FULL). Besides, APRI increased WUE by 9.63% (95%CI: 7.65–11.64%), 4.56% (1.38–7.84%), 11.18% (4.21–18.62%), when the deficit proportion was 80–100%, 60–80% and 40–60%, respectively (Fig. 4f).

3.3.2. APRI by DI

Compared with DI, APRI of different irrigation methods, deficit stages and proportions significantly increased WUE, while the changes of yield were inconsistent (Fig. 5). Yields of drip and furrow irrigation were both decreased (−5.94% and −2.88%, respectively); thereafter, the percentage change of yield in response to sprinkler irrigation increased, which may be due to the small sample size (n = 3) (Fig. 5a). The WUE of drip and furrow irrigation was increased 4.17% and 5.41%, respectively, but the results were not significant different (Fig. 5b).

In terms of deficit stage, APRI increased yield and WUE relative to DI when the deficit occurred at the reproductive stage. Specifically, the yield was increased by 4.29% (95%CI: 0.5–8.23%) and WUE was increased by 2.35% (95%CI: −3.46–8.5%). However, the result of WUE was not significant different because the 95%CI included 0 (Fig. 5c, d). No matter that deficit stage occurred at reproductive stage or throughout the season, the difference between them was not significant (Fig. 5d).

Compared with DI, the reduction of yield under APRI was positively correlated with deficit proportions (Fig. 5e). Namely, APRI decreased yield by 4.7%, 5.72% and 11.54% when the deficit proportions were 80–100%, 60–80% and 40–60%, respectively. Nevertheless, when the deficit proportions were 40–60% and 60–80%, the difference between APRI and DI was not significant, because the 95% confidence internal of them included 0 (Fig. 5e). The largest WUE (47.36%) occurred when the deficit proportion was 40–60%, while the smallest WUE (−15.48%) occurred at 60–80% (Fig. 5f). However, the sample size of the deficit proportion of 40–60% was small, so the results need to be further verified.

3.3.3. DI by FULL

Under different irrigation methods, deficit stages and deficit

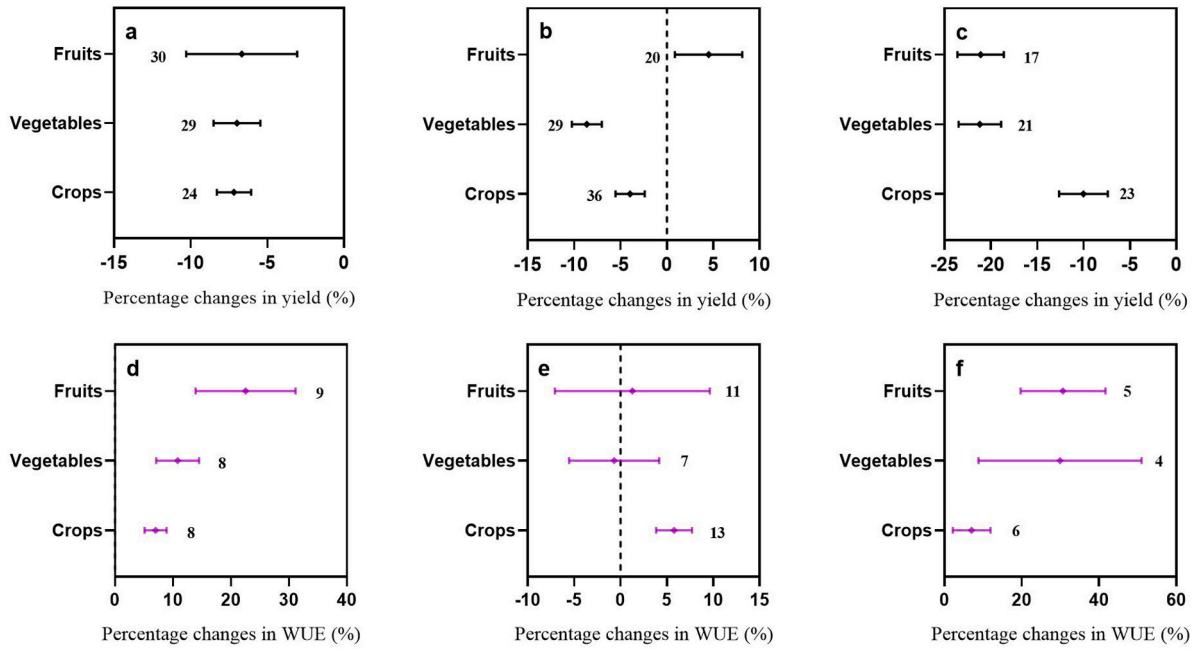


Fig. 3. The effects of different irrigation methods (APRI-to-FULL (a, d), APRI-to-DI (b, e), DI-to-FULL (c, f)) on yield and WUE, as a function of agricultural species groups. Mean effect size = 0 (vertical dotted line) indicates no effect, mean effect > 0 indicates yield or WUE advantage of APRI over FULL, APRI over DI, DI over FULL and mean effect size < 0 indicates yield or WUE advantage of FULL over APRI, DI over APRI, FULL over DI. Number beside the every bar represents the sample size.

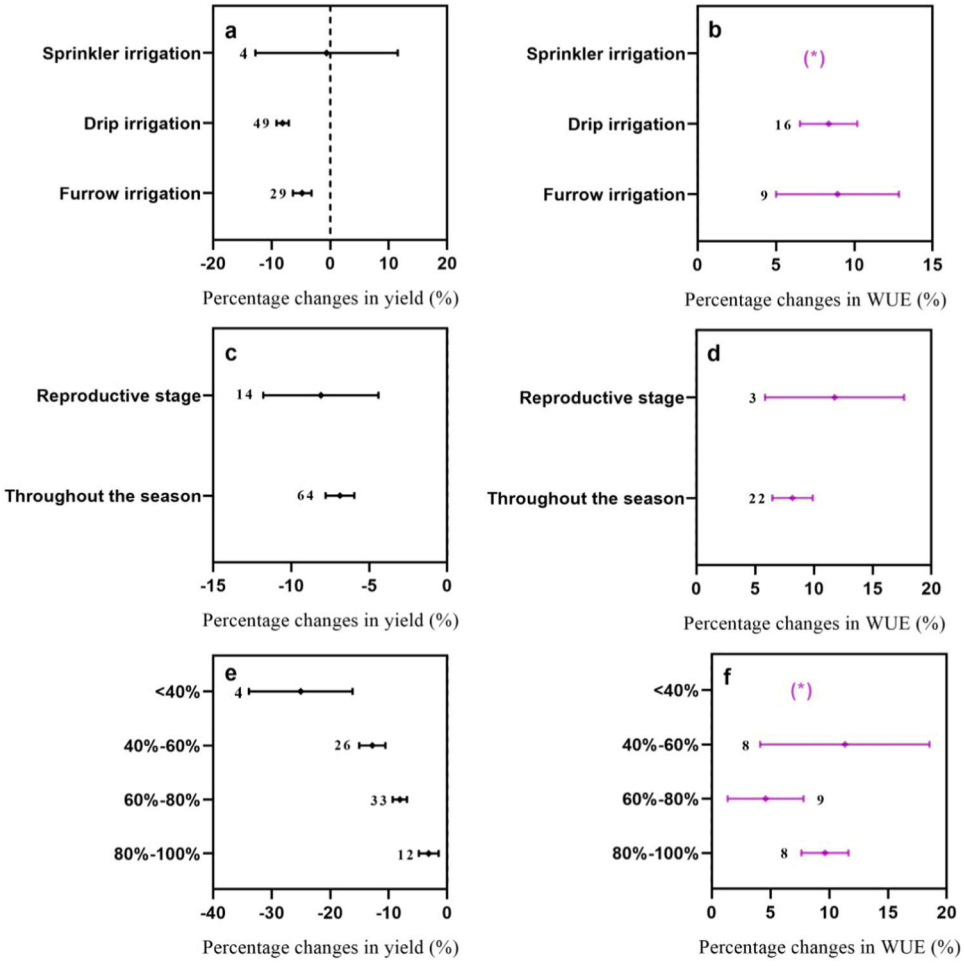


Fig. 4. The effects of APRI on yield and WUE as a percentage change of FULL for different irrigation methods (a, b), deficit stages (c, d) and deficit proportions (e, f). Mean effect and 95% confidence interval are shown. The asterisk (*) represents no comparison in this subgroup. Mean effect size = 0 (vertical dotted line) indicated no effect, mean effect > 0 indicates yield or WUE advantage of APRI over FULL and mean effect size < 0 indicates yield or WUE advantage of FULL over APRI. Number beside the every bar represents the sample size.

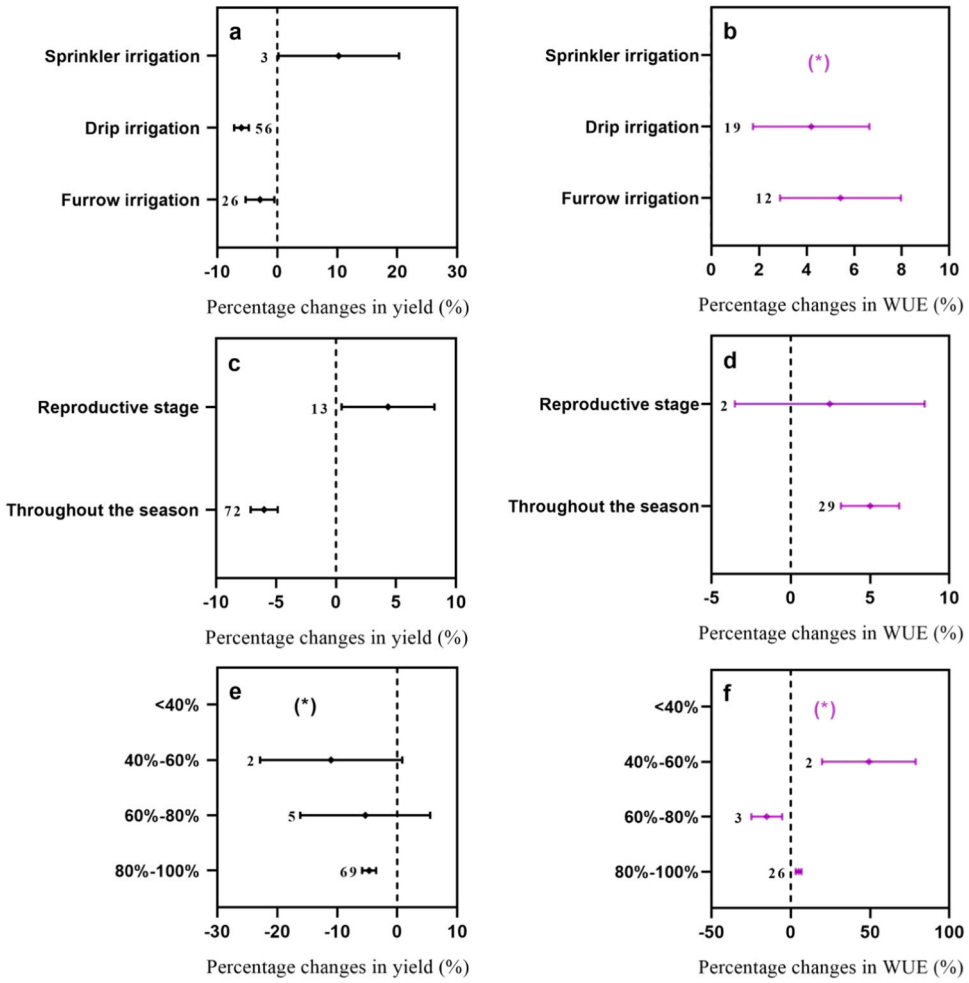


Fig. 5. The effects of APRI on yield and WUE as a percentage change of DI for different irrigation methods (a, b), deficit stages (c, d) and deficit proportions (e, f). Mean effect and 95% confidence interval are shown. Mean effect size = 0 (vertical dotted line) indicates no effect, mean effect > 0 indicates yield or WUE advantage of APRI over DI and mean effect size < 0 indicates yield or WUE advantage of DI over APRI. Number beside the every bar represents the sample size.

proportions, yield and WUE of DI were affected to varying degrees relative to FULL (Fig. 6). The yields and WUE for all irrigation methods considered in this meta-analysis, except sprinkler irrigation, were all significantly different from 0, but the effect size for yield was not significant due to the small size (n = 3) when using DI under sprinkler irrigation (Fig. 6a, b). The percentage change in yield under furrow irrigation (−9.6%) was significantly greater than that under drip irrigation (−22.54%), but it was slightly greater than that under sprinkler irrigation (−10.17%) (Fig. 6a). The WUE of furrow irrigation (8.11%) was lower than that of drip irrigation (18.3%), but their results were not significant (Fig. 6b).

With regard to the influence of deficit stage, the reduction of yield of DI occurred at reproductive stage (−20.54%, 95%CI: −23.02% to −17.98%) was greater than that occurred throughout the season (−16.02%, 95%CI: −17.76–14.24%), and the differences were significant (Fig. 6c). However, the WUE under water deficit throughout the season was significantly higher than that at reproductive stage, and increased WUE by 20.76% (95%CI: 12.02–27.72%) (throughout the season) and 5.85% (95%CI: 2.00–14.18%) (reproductive stage) (Fig. 6d).

The percentage changes of yield and WUE under DI were also affected by deficit proportion, relative to FULL (Fig. 6e, f). Specifically, the yield was negatively related to deficit proportion, and yield were decreased by 6.43% (80–100%), 15.34% (60–80%) and 21.49% (40–60%), respectively (Fig. 6e). The WUE of deficit proportion at 40–60% (13.79%) was marginally higher than that at 60–80% (11.96%), but these results were not significantly different (Fig. 6f).

3.4. Yield and WUE response of soil properties to different irrigation methods

3.4.1. APRI by FULL

Regardless of soil properties, APRI decreased yield but increased WUE compared with FULL except for WUE for soil bulk density < 1.35 g cm^{-3}, with varying degrees of success (Fig. 7). APRI decreased yield in clay, sandy clay, clay loam, sandy clay loam, loam, sandy loam and sand by 2.12%, 4.6%, 4.86%, 28.64%, 7.41%, 3.47% and 9.14% relative to FULL, respectively. The increases in WUE of APRI relative to FULL were 6.65%, 20.56%, 7.18% and 11.52% in sandy clay, clay loam, loam and sandy loam, respectively (Fig. 7a and b).

The relative change of APRI on yield was negatively correlated to bulk density, but the trend was opposite for WUE (Fig. 7c, d). The reduction of yield for soils with bulk density < 1.35 g cm^{-3} was 17.10% (95%CI: −19.49% to −14.64%) lower than that of soils with bulk density ≥ 1.35 g cm^{-3} (−5.42%, 95%CI: −6.44–4.38%), and the difference was significant (Fig. 7c). However, WUE for soils with bulk density ≥ 1.35 g cm^{-3} was increased by 8.01% (95%CI: 6.24–9.82%), which was significantly higher than that for soils with bulk density < 1.35 g cm^{-3} (−30.97%, 95%CI: −43.01–16.37%) (Fig. 7d).

With regard to the influence of soil pH on APRI, APRI (compared to FULL) decreased yield by 1.45% (95%CI: −4.68–1.89%) and 14.18% (−17.07% to −11.19%) for soils with pH < 7 and ≥ 7, respectively, and

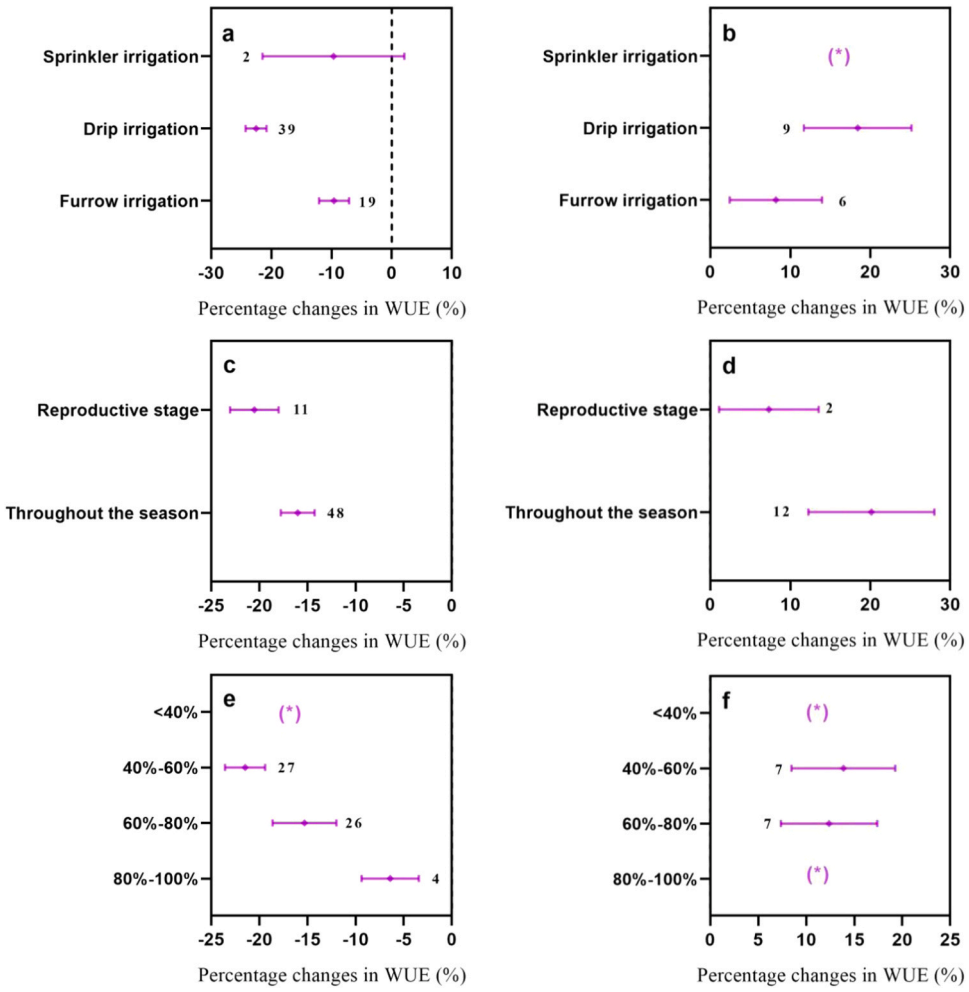


Fig. 6. The effects of DI on yield and WUE as a percentage change of FULL for different irrigation methods (a, b), deficit stages (c, d) and deficit proportions (e, f). Mean effect and 95% confidence interval are shown. Mean effect size = 0 (vertical dotted line) indicated no effect, mean effect > 0 indicates yield or WUE advantage of DI over FULL and mean effect size < 0 indicates yield or WUE advantage of FULL over DI. Number beside the every bar represents the sample size.

the reduction of yield for soils with pH < 7 was not significant (Fig. 6e).

3.4.2. APRI by DI

The relative changes of yield and WUE of APRI relative to DI was affected by soil properties (soil texture, bulk density and pH) (Fig. 8). Except for clay and sandy loam, other soil textures decreased yield to varying degrees. Specifically, APRI decreased yields in sandy clay, silt clay loam, clay loam, sandy clay loam, silt loam, loam and sand by 5.29%, 1.09%, 21.89%, 0.55%, 4.46%, 1.96% and 1.54%, respectively (Fig. 8a). In contrast, except for clay loam (−26.93%), APRI increased WUE by 4.94%, 4.04%, 2.35% and 7.56% when soil textures were clay, sandy clay, loam and sandy loam. Apart from clay and loam, the results were significant (Fig. 8d).

Regardless of soils with bulk density ≥ 1.35 g cm^{-3} or < 1.35 g cm^{-3}, APRI decreased yield significantly relative to DI, and the reduction of yield for soil with bulk density ≥ 1.35 g cm^{-3} (3.52%) was significantly higher than that with bulk density < 1.35 g cm^{-3} (15.73%) (Fig. 8b).

The relative changes of yield between the two soil pH were opposite, but these results were not remarkably different between the two different irrigation regimes because the 95%CI of soil pH included 0. Specifically, the decrease of yield for soil with pH < 7 was 3% (95%CI: −6.52–0.65%) and the increase for soils with pH ≥ 7 was 3.37% (−0.59–7.5%) (Fig. 8c).

3.4.3. DI by FULL

Soil texture and bulk density decreased yield but increased WUE relative to FULL with varying degrees of success (Fig. 9). The largest decrease occurred in sandy loam (25.37%), while the smallest decrease occurred in loam (2.67%) (Fig. 9a). The relative changes in WUE under DI compared with FULL were 43.24%, 5.85%, 9.94% and 4.99% for clay loam, loam, sandy clay and sandy loam, respectively (Fig. 9b).

The smallest decrease in yield and the largest increase in WUE under DI occurred in soils with bulk density < 1.35 g cm^{-3}, with values of −13.12% (95%CI: −16.49–9.61%) and 28.77% (95%CI: 9.40–51.57%) respectively, which were higher than those in soils with bulk density ≥ 1.35 g cm^{-3} (13.65% and 6.14%, respectively). However, the results between these two soil bulk densities were not significant (Fig. 9c, d).

3.5. Yield and WUE response of meteorological factors to different irrigation methods

3.5.1. APRI by FULL

Regarding to the response of yield and WUE to different meteorological factors (precipitation and temperature) under APRI compared to FULL, the two seasonal precipitation levels during both significantly increased WUE but caused a significant reduction in yield. As for the average annual temperature, it significantly increased WUE but not yield (Fig. 10).

Seasonal precipitation < 200 mm caused a larger decrease in yield

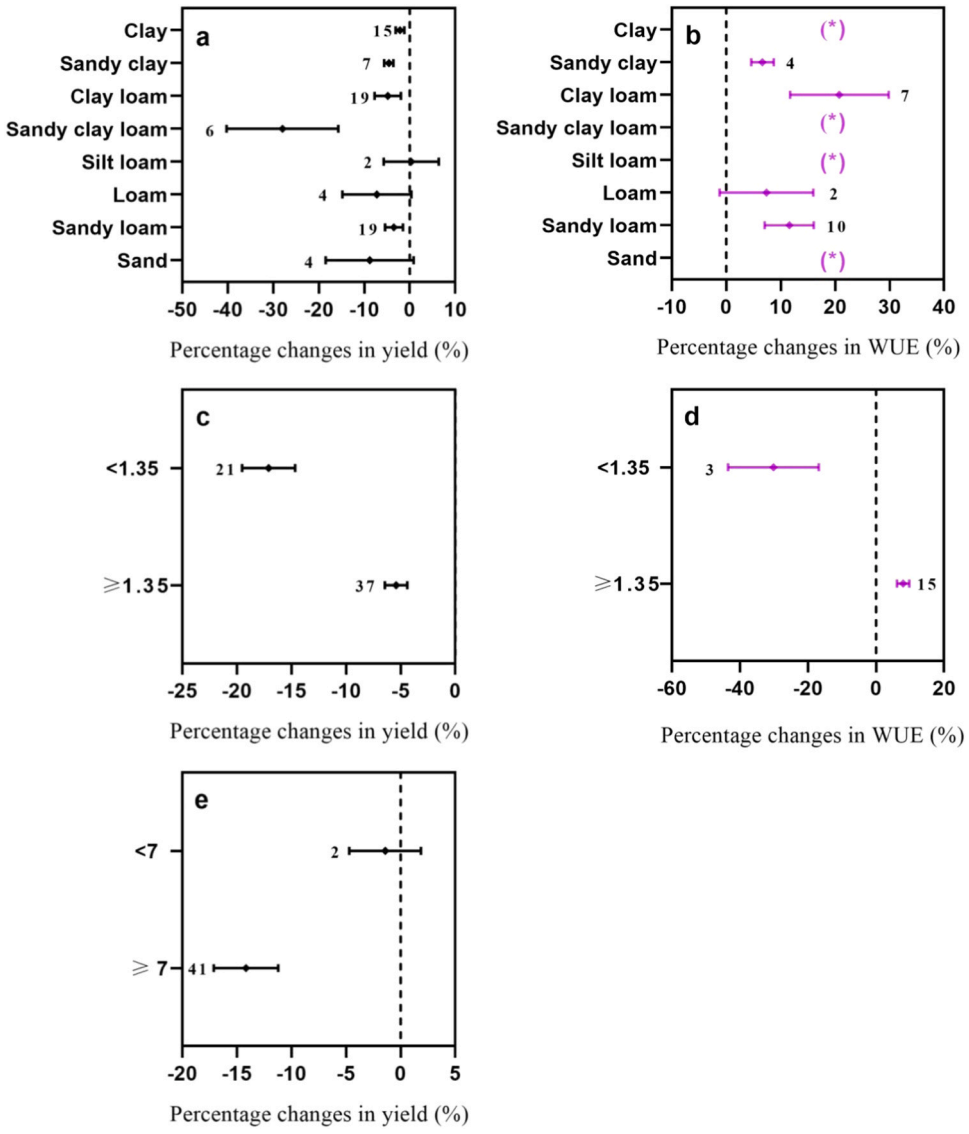


Fig. 7. The effects of APRI on yield and WUE as a percentage change of FULL for different soil texture (a, b), bulk density (c, d) and pH (e). Mean effect and 95% confidence interval are shown. Mean effect size = 0 (vertical dotted line) indicates no effect, mean effect > 0 indicates yield or WUE advantage of APRI over FULL and mean effect size < 0 indicates yield or WUE advantage of FULL over APRI. Number beside the every bar represents the sample size.

by −5.59% (95%CI: −7.19% to −3.96%), but no significant difference was observed for seasonal precipitation > 200 mm (−4.83%, 95%CI: −5.8% to −3.84%). However, seasonal precipitation < 200 mm significantly increased WUE by 12.49% (95%CI: 8.67–16.44%), which was significantly higher than seasonal precipitation > 200 mm (5.62%, 95% CI: 5.08–6.16%) (Fig. 10a, b).

The percentage increase in yield for average annual temperature < 9 7¯C was 10.4% (95%CI: 6.4–14.6%) and 1.08% (95%CI: −1.41–3.63%), but the difference was not significant because of the 95% CI containing 0. Average annual temperature > 9 °C increased WUE but not yield. The percentage change was a 5.7% increase in WUE but a 6.61% decrease in yield (Fig. 10c, d).

3.5.2. APRI by DI

The relative changes of yield and WUE under APRI to DI were also influenced by the seasonal precipitation and average annual temperature (Fig. 11). The yield under APRI in regions with seasonal precipitation > 200 mm (−4.83%, 95%CI: −5.80% to −3.84%) was larger than that < 200 mm (−5.59%, 95%CI: −7.19%to −3.96%) compared to DI (Fig. 11a). The percentage changes of yield under APRI at different temperatures were either significantly higher than or not significantly different from that under DI (Fig. 11c). APRI caused higher WUE than DI regardless of precipitation levels during the growing season (Fig. 11b). Although there were positive responses of WUE at different temperatures, the difference was not significant (Fig. 11d).

3.5.3. DI by FULL

The effect of DI on yield increased with increasing seasonal precipitation amount (Fig. 11a). The percentage change of greater seasonal precipitation (> 200 mm) (−7.6%) was significantly higher than that of low precipitation (< 200 mm) (−22.3%). Although higher percentage change was found for seasonal precipitation > 200 mm (5.0%), no significant difference was observed between the two precipitation levels. Moreover, the reduction of WUE for low seasonal precipitation (< 200 mm) under DI was not significantly different from that under FULL (Fig. 11b). The percentage decrease of yield and WUE were 12.7% and 7.0% in regions with higher average annual temperature (> 9 °C), respectively (Fig. 11c, d).

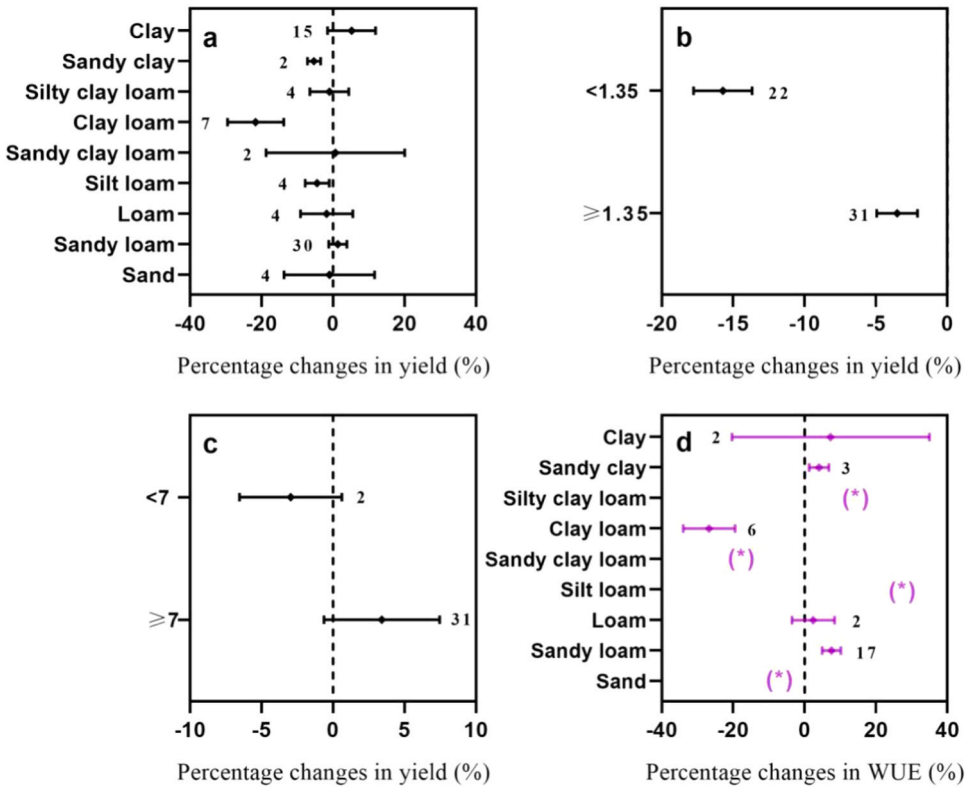


Fig. 8. The effects of APRI on yield and WUE as a percentage change of DI for different soil texture (a, b), bulk density (c) and pH (d). Mean effect and 95% confidence interval are shown. Mean effect size = 0 (vertical dotted line) indicated no effect, mean effect > 0 indicates yield or WUE advantage of APRI over DI and mean effect size < 0 indicates yield or WUE advantage of DI over APRI. Number beside the every bar represents the sample size.

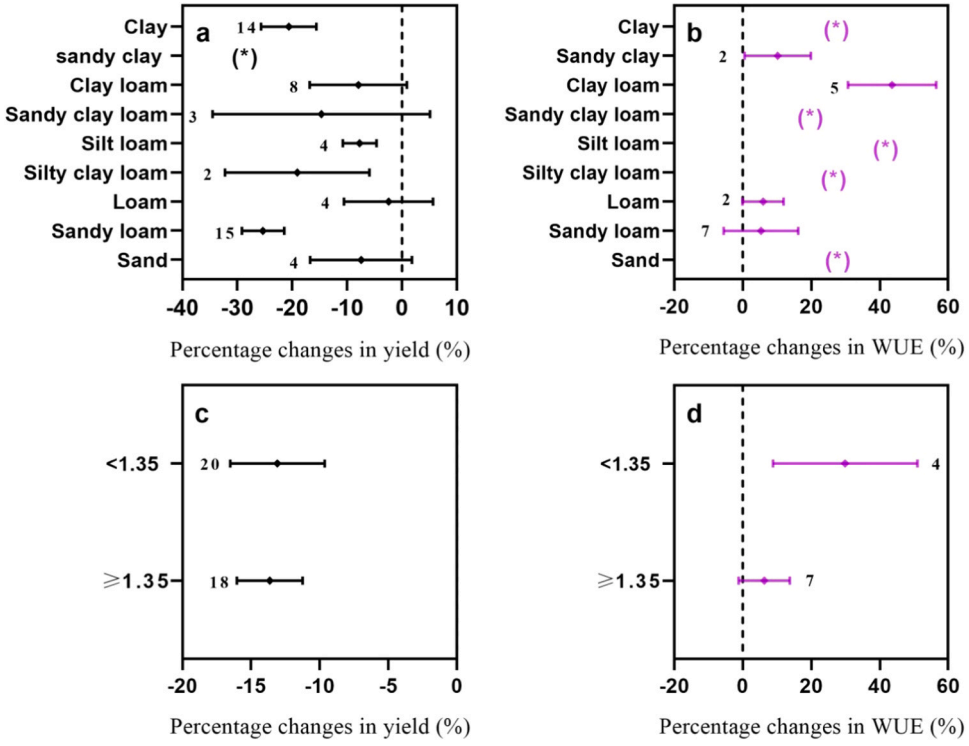


Fig. 9. The effects of DI on yield and WUE as a percentage change of FULL for different soil texture (a, b) and bulk density (c, d). Mean effect and 95% confidence interval are shown. Mean effect size = 0 (vertical dotted line) indicated no effect, mean effect > 0 indicates yield or WUE advantage of DI over FULL and mean effect size < 0 indicates yield or WUE advantage of FULL over DI. Number beside the every bar represents the sample size.

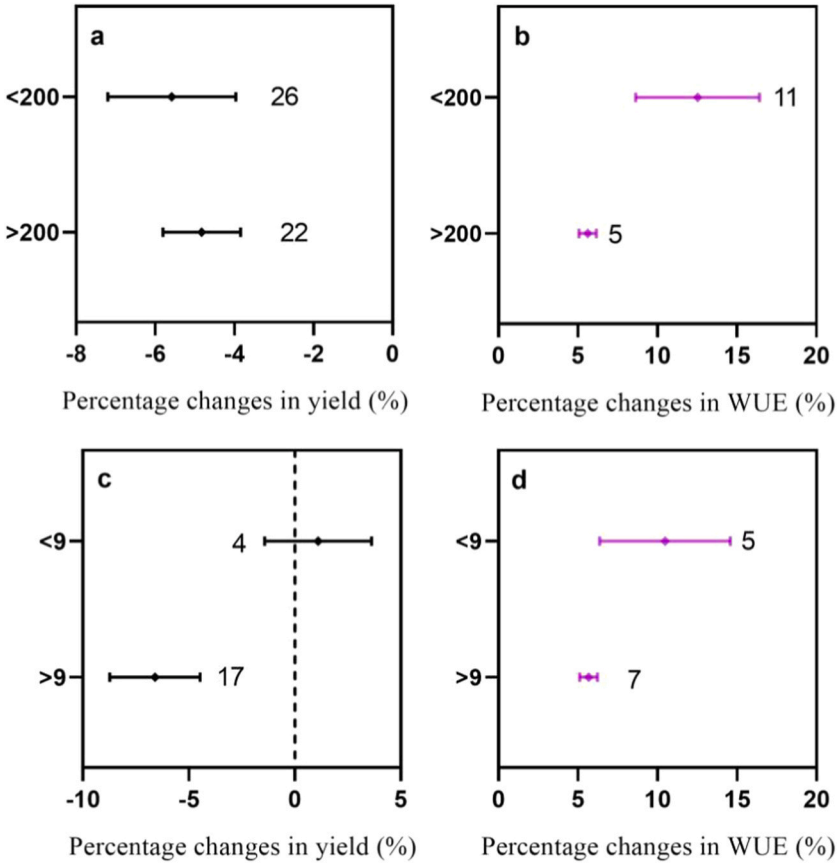


Fig. 10. The effects of APRI on yield and WUE as a percentage change of FULL for different seasonal precipitation amounts (a, b) and average annual temperatures (c, d). Mean effect and 95% confidence interval are shown. Mean effect size = 0 (vertical dotted line) indicates no effect, mean effect > 0 indicates yield or WUE advantage of APRI over FULL and mean effect size < 0 indicates yield or WUE advantage of FULL over APRI. Number beside the every bar represented the sample size.

4. Discussion

4.1. Yield and WUE of agricultural species groups

Compared with FULL, yield under APRI and DI significantly decreased, and the degree varied significantly among agricultural species groups (Fig. 3a, c), suggesting that different agricultural species groups had various responses to water deficit (McCarthy, 1999; Pérez-Pérez et al., 2012), thereby these differences led to yield differences (Costa et al., 2007; Steduto et al., 2012). Wang et al. (2018, 2019) concluded that the irrigation level significantly affected potato yield and seed cotton yield. Furthermore, compared with FULL, the significant reduction of yield under APRI and DI was probably due to the deficit stage, including the reproductive stage (Adu et al., 2018). The decrease of yield under APRI were similar for fruits, vegetables and crops, whereas the WUE of fruits was increased most among the three agricultural species groups compared with FULL (Fig. 3a, d). Besides, when compared with DI, both yield and WUE of fruits increased under APRI (Fig. 3b, e). Therefore, APRI was more conducive to fruit trees compared with other agricultural species. The result is similar to those reported by Du et al. (2005), who revealed that fruit trees might be more suitable for the application of APRI technique because of their deeper root system and wide spacing. Similar results were also demonstrated by Du et al. (2008b), Zegbe and Behboudian (2008) and dos Santos et al. (2003). The yield reduction under DI was lager for fruits and vegetables than crops compared with FULL. However, the trend of WUE was opposite (Fig. 3c, f). Therefore, the choice of deficit irrigation depends to a large extent on economically weighting the loss of yield and the benefits of saving irrigation water for different agricultural species.

The comparison results of crop yield and WUE under APRI and DI were inconsistent (Fig. 3b, e). For fruits, APRI significantly increased yields by 4.46%, while vegetables and crops significantly decreased by 8.64% and 3.97%, respectively. The results may be due to the differences in applied water and the duration of deficit under APRI and DI. After all, different agricultural species and cultivars have different response to water stress (Goodwin and Macrae, 1990; McCarthy, 1999). There was no difference or significant increase in WUE between APRI and DI, which agreed with the finding of Peng (2012), who found PRD improved the drought resistance of crops and saved water, compared with deficit irrigation. WUE under DI and APRI were significantly increased among all crops (Fig. 3d, f). The results are consistent with that of Du et al. (2010, 2017) and Sadras (2009).

4.2. Yield and WUE of irrigation managements

In this meta-analysis, we found that the response of crop yields and WUE to APRI and DI varied significantly in response to irrigation management practices relative to FULL, including irrigation strategy, deficit stage and proportion (Figs. 4–6). Regardless of DI or APRI (compared with FULL) or APRI compared with DI, changes of yield of drip irrigation showed a larger decrease compared to furrow irrigation (Figs. 4a–6a). This may be because the reduced irrigation water used in furrow irrigation was able to improve the oxygen content in soils and reduced nutrient loss caused by anaerobic respiration, while drip irrigation has been verified as a means of regular and even water supply (Cetin and Bilgel, 2002; Yu et al., 2020). As for drip irrigation, the water condition of crops under drip irrigation was probably inherently better than that of furrow irrigation, so the reduced water supply under drip irrigation resulted in a reduction in yield (Camp, 1998). However, WUE of pairwise comparisons of irrigation methods were all significantly increased compared to the control treatment (Figs. 4b–6b). This can be largely attributed to the reduced evapotranspiration (Sezen and Yazar, 2006).

With respect to the deficit stage, the greater reduction in yield

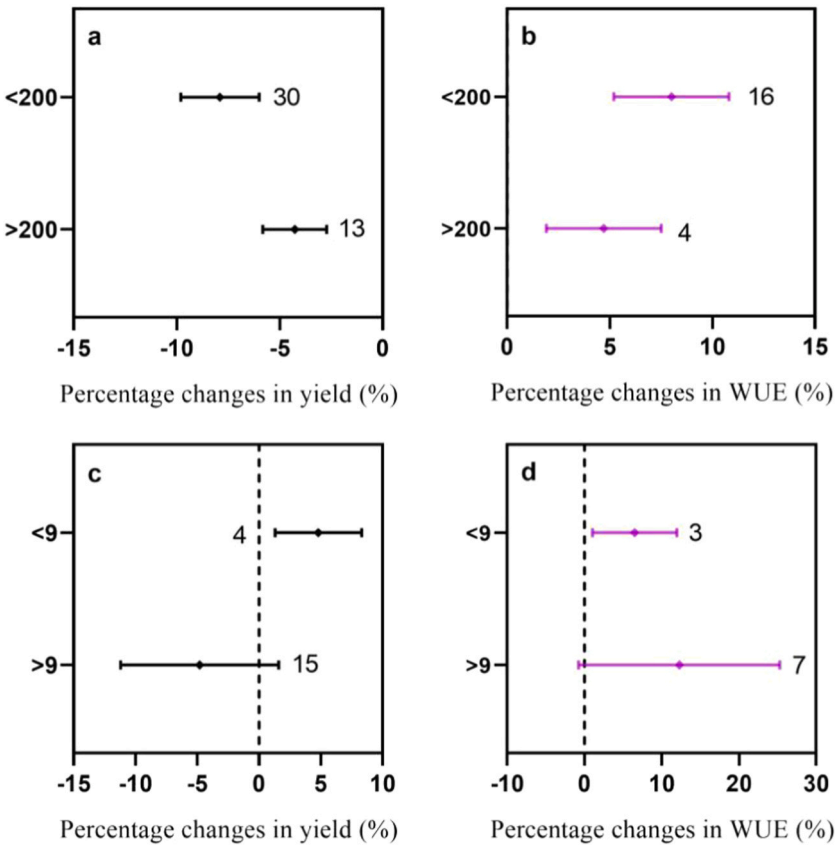


Fig. 11. The effects of APRI on yield and WUE as a percentage change of DI for different seasonal precipitation amounts (a, b) and average annual temperatures (c, d). Mean effect and 95% confidence interval are shown. Mean effect size = 0 (vertical dotted line) indicates no effect, mean effect > 0 indicates yield or WUE advantage of APRI over DI and mean effect size < 0 indicates yield or WUE advantage of DI over APRI. Number beside the every bar represented the sample size.

occurred when APRI and DI were applied at the reproductive stage, which is in agreement with previous studies (Cheng et al., 2018; Jin et al., 2020; Zhang et al., 2019). One possible reason is that the water deficit at the reproductive stage may cause early wilting of the lower leaves of crops, which reduce the radiation intercepted by crops, leading to a decrease in crop yields (Cheng et al., 2018). Another probable reason may be due to the water stress occurred at flowering (Robertson and Giunta, 1994), which significantly inhibits the formation of flowers and ultimately reduces the number of grains (Grewal, 2010). The results also revealed that deficit irrigation throughout the season showed a moderate but not significant higher WUE, which is in accordance with the results of Yu et al. (2020). The possible reason is that the water deficit at the vegetative stage is too early to influence yield but saves water to a certain degree (Lu et al., 2019).

The responses of yield to different irrigation managements not only depend on the irrigation strategy and deficit stage but also on the deficit proportion. We found APRI and DI reduced yield significantly when reduced the amount of irrigation water at each level in comparison to FULL, which contradicts to the previous results that APRI can save 30% of full irrigation water, improve WUE and do not significantly reduce yield over full irrigation (Du et al., 2008b; Fu et al., 2017; Giuliani et al., 2017; Shahnazari et al., 2008). The difference may be due to the different deficit stage and the duration of wet and dry cycling and intensity (Helyes and Varga, 1994; Nuruddin et al., 2003; Zegbe et al., 2006). In this meta-analysis, when the irrigation proportion of APRI and DI was within 80–100% FULL, the duration of the wet and dry cycling under field conditions considered in this meta-analysis was mostly longer than 14 d, and it influenced the intensity of the APRI and DI responses (Ahmadi et al., 2010; Yactayo et al., 2013). Another reason for this could be the deficit stage. In this meta-analysis, deficit was applied either at reproductive or throughout the season, and reproductive stage was the stage of yield formation (Chen et al., 2014). Therefore, the effect of reduced water supply on crop yields was relatively large.

4.3. Yield and WUE of soil properties

Generally, for most soil properties, yield of APRI and DI (compared with FULL) decreased, while WUE increased obviously (Figs. 7a, b and 9a, b). However, there was no significantly difference in yield between APRI and DI (Fig. 8a). As for the former, yield in sandy clay loam was most negative and WUE in clay loam was most positive (Fig. 7a, b). These results were in accordance with the findings of Adu et al. (2018), who found the better yield in sandy clay loam (referring to the relative yield of FI to DI and PRDI) over clay silt loam. This may be due to the poor water retention of sandy soil, where plants may soon suffer from the water stress at a water deficit stage. Therefore, the largest reduction in yield in sandy soil was not entirely unexpected (Kirda, 2002; Zhang et al., 2004). Besides, the texture characteristics of soils also affected the rooting and water extraction capabilities of crops, thereby limiting the ability of plants to resist water stress (Tolk et al., 1999; Zhang et al., 2004). WUE of APRI by FULL in clay loam was most positive (Fig. 7b). Crops grow in finer soils may have enough time to adapt to the lower soil water content, and to adjust the drought response and reduce the negative effect of water deficit (Kirda, 2002). Simultaneously, it also showed that water-saving irrigation strategies did not necessarily have significant results in all soils.

The performance of yield in response to APRI by FULL and APRI by DI for soil bulk density ≥ 1.35 g cm^{-3} was better than that for soil bulk density < 1.35 g cm^{-3} (Figs. 7c, and 8c). The reason for this result may be due to that high soil bulk density can result in soil hypoxia or anoxia due to the increased soil compaction and reduced nutrient uptake, elongation of roots and soil aeration (Wolkowski and Lowery, 2008).

Obviously, these factors significantly limited the growth of crops (Ityel et al., 2014). APRI maintained and controlled the active root layer of the soil to alternately appear in the dry area of the vertical or horizontal profile, which significantly improved the soil aeration and elongation of roots (Kang et al., 1997).

The yield of APRI (compared with FULL) for soil pH < 7 was better than that for soil pH ≥ 7 (Fig. 7e). This was probably because the drought caused by APRI promoted the aerobic respiration (Du et al., 2018), which led to the volatilization of ammonia in the root soil (Carrijo et al., 2017), and the decrease in soil nitrogen uptake and possibly yield. Compared with DI, APRI increased yield for soil pH ≥ 7, while yield was decreased for soil pH < 7, but this result was not significant because the 95% confidence internal of soil pH included 0 (Fig. 8c).

4.4. Yield and WUE of meteorological factors

Precipitation and temperature are widely recognized as two main factors influencing yield and WUE (Mo et al. 2018; Yu et al. 2020; Wang et al. 2020). Our results indicated that the percentage increase of WUE under APRI in regions with higher seasonal precipitation (> 200 mm) was significant lower than that with low precipitation (< 200 mm) (Fig. 10b). The possible reason can be that higher precipitation caused greater ET (Zheng et al., 2019). The yield and WUE in regions with average annual temperature < 9 °C were higher than those with average annual temperature > 9 °C (Fig. 10c, d). It was mainly due to that agricultural species planted in regions with low temperature (< 9 °C) could extend the growing season length (Zhang et al., 2018) and thus produced large yield and WUE.

Greater yield under APRI were easier to achieve in regions with higher seasonal precipitation (> 200 mm), while the percentage increase of WUE was larger in low precipitation environment (< 200 mm) compared with DI (Fig. 11a, b). The result may be greatly attributed to that APRI promoted the lateral root growth (Cheng et al. 2018, 2021), which was beneficial to the utilization of soil water and fertilization in low precipitation environment (< 200 mm) (Wang et al. 2010). Besides, APRI reduced the soil water loss by evaporation relative to DI, thereby increasing yield and WUE to some degree.

The yield and WUE under DI showed an increasing trend with increasing seasonal precipitation compared to FULL (Fig. 12a, b). It indicated that higher yield and WUE under DI could be more likely to achieve in regions with higher seasonal precipitation (> 200 mm). One possible reason can be that DI could avoid the excessive soil saturation in higher precipitation environment which suppressed the root respiration (Iqbal et al., 2011), and affected crop water and fertilizer absorption, thereby hindering yield and WUE (Xu et al., 2016).

4.5. Limitations and implications

The meta-analysis evaluated the effects of different irrigation, meteorological and soil factors on yield and WUE of seven agricultural species under APRI, DI and FULL. However, because of the insufficient data in WUE, the effects of soil pH on the response of WUE to APRI, DI and FULL could not be side-by-side compared. In addition, the response of yield to DI compared to FULL for different soil textures was also not discussed. Therefore, these missing comparisons need to be done in the future. In this meta-analysis, we only investigated the responses of maize, wheat, cotton, potato, tomato, apple and grape to APRI, DI and FULL. Indeed, some other varieties also met these criteria, but the data were collected from only one or two published papers. These seven

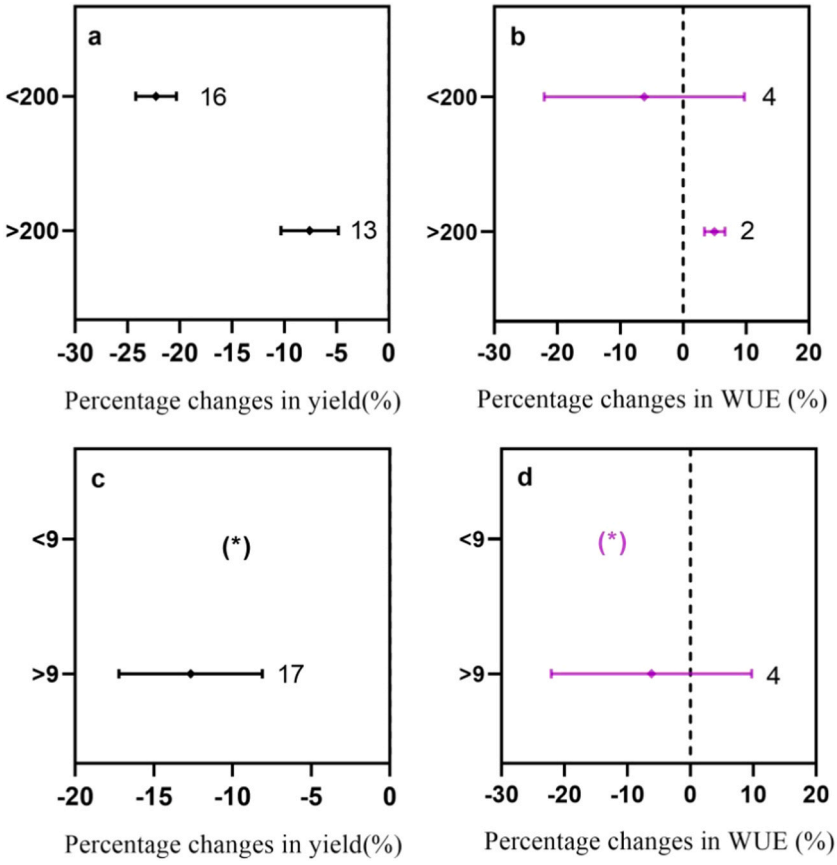


Fig. 12. The effects of DI on yield and WUE as a percentage change of FULL for different seasonal precipitation amounts (a, b) and average annual temperatures (c, d). Mean effect and 95% confidence interval are shown. Mean effect size = 0 (vertical dotted line) indicates no effect, mean effect > 0 indicates yield or WUE advantage of DI over FULL and mean effect size < 0 indicates yield or WUE advantage of FULL over DI. Number beside the every bar represented the sample size.

varieties were studied relatively more than other varieties among the collected literature. This meta-analysis analyzed the effects of APRI, DI, FULL on yield and WUE considering agricultural species group, soil texture, bulk density, pH, irrigation method, the amount of water applied and the period when APRI or DI irrigation was undertaken. However, the frequency of irrigation and fertilization also affected the results. Thus, more factors need to be analyzed to investigate the responses of yield and WUE.

5. Conclusions

This meta-analysis analyzed the percentage changes of yield and WUE of seven agricultural species under APRI, DI and FULL under different irrigation managements, soil properties and meteorological factors. Overall, yield under APRI and DI were remarkably lower than those under FULL, while the corresponding WUE were significantly higher. There was no significant difference in yield between APRI and DI, but WUE under APRI was higher. Compared with FULL, APRI performed better in fruits with irrigation amount of 80–100% FULL, silt loam, loam and sand, soils with high bulk density (≥ 1.35 g cm^{-3}) and furrow irrigation. Besides, the response of APRI was better in regions with low seasonal precipitation (< 200 mm) and low average annual temperature (< 9 °C). Compared with FULL, DI performed better for clay loam, low bulk density (< 1.35 g cm^{-3}) and higher seasonal precipitation (> 200 mm). Compared with DI, fruits with furrow irrigation and soil with clay and sandy loam were better under APRI. APRI in regions with low average annual temperature (< 9 °C) was more likely to increase both yield and WUE. Therefore, it is critical to weigh the cost of yield penalty, the economic benefit of saving water and the cost of implementing irrigation methods before choosing various irrigation methods.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.agwat.2021.106771.

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